CENTRE: A Bioconductor package for cell-type specific enhancer-target prediction

Sara Lopez Ruiz de Vargas[1], Trisevgeni Rapakoulia[1,2], Persia Akbari-Omgba[1], Verena Laupert[1,3], Martin Vingron[1]

[1]Max Planck Institute for Molecular Genetics, Computational Molecular Biology, Berlin, Germany
[2]Present Adress Present address: GlaxoSmithKline plc, Gunnels Wood Road, Stevenage SG1 2NY, United Kingdom
[3] Present address: Bayer AG, Muellerstrasse 178, Berlin 13353, Germany

MAX PLANCK INSTITUTE
FOR MOLECULAR GENETICS

Bioconductor
OPEN SOURCE SOFTWARE FOR BIOINFORMATICS

ABSTRACT

Enhancers are non-coding DNA regions ranging from 100bp to 1000bp in size. They serve as a landing space for Transcription Factors and interact with promoter regions to regulate gene expression dynamically. The configuration of active enhancer-promoter interactions is cell-type and condition-specific. Identifying active enhancer-promoter pairs is crucial to understanding gene regulation, phenotypes, and diseases. Up to now, several computational methods have been developed to predict enhancer gene interactions, but they require many epigenomic and transcriptomic experimental assays to generate cell-type-specific predictions. Thus, inferring enhancer gene interactions becomes laborious and costly, especially when looking for cell type (CT) specific contacts.

Recently, we introduced the workflow of CENTRE[1], aimed at predicting CT-specific enhancer-target interactions with minimal experimental input. CENTRE uses an XGBoost[2] classifier to predict enhancer target interactions in a CT-specific manner, relying solely on gene expression and ChIP-seq data for three histone modifications pertinent to the CT of interest. Leveraging available datasets, CENTRE combines cell-type-agnostic statistics with CT-specific features. For enhanced software accessibility, we now implement the CENTRE workflow as a Bioconductor package and introduce a novel Hi-C module, enabling users to incorporate Hi-C data seamlessly without needing to provide it themselves. CENTRE has undergone extensive validation across diverse datasets and cell types, consistently demonstrating comparable or superior performance to existing algorithms reliant on extensive experimental data.

IMPACT

We present CENTRE as a tool for enhancer-target prediction. CENTRE requires little input data from the user and is implemented as a Bioconductor package, enabling easy routine use and seamless integration with downstream analysis.

CENTRE performs comparable to or better than state-of-the-art algorithms.

FRAMEWORK

Step 1. CENTRE will extract all the enhancers within 500kb of the input target genes (ET pairs).

Step 2. Compute the generic features for all ET pairs. All generic features except genomic distance are precomputed for each ET pair and stored in PrecomputedData.db.

Step 3. Compute the cell type-specific features from the user-given cell type-specific experiments. This step uses CRUP[3] scores and RNA-seq to calculate enhancer and promoter activity of genomic regions.

Step 4. Feed all features to the pre-trained XGBoost[2] classifier and return a probability of interaction assigned to the pairs. Pairs with probabilities higher than 0.5 are labeled as interaction.

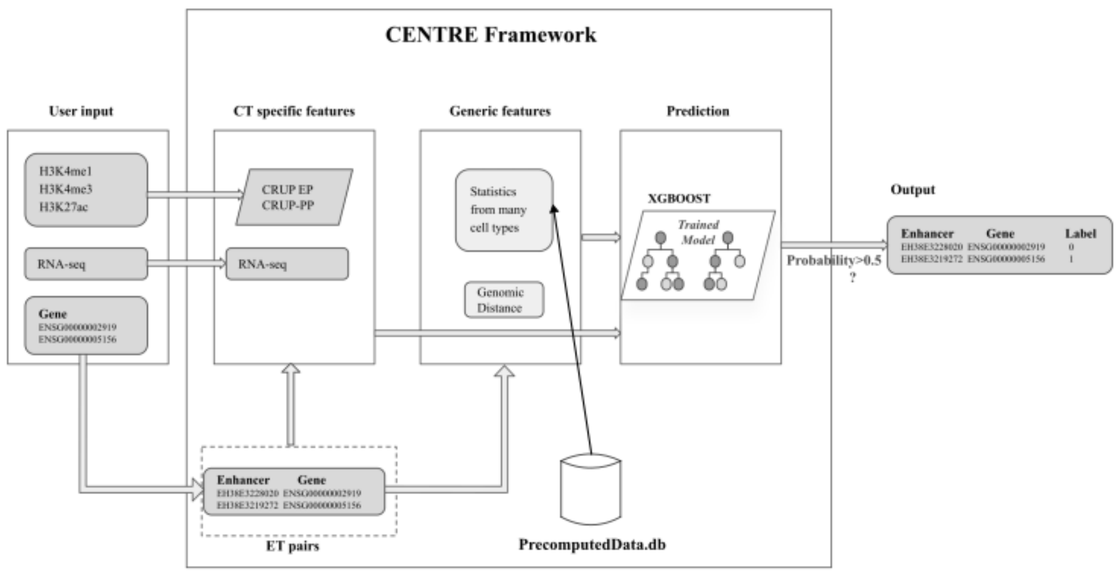

Fig. 1- Outline of CENTRE framework: the user provides target genes of interest (ENSEMBL id) with CT-specific RNA-seq (TSV format gene quantifications) and H3K27ac, H3K4me1, and H3K4me3 ChIP-seq data (BAM format).

R-PACKAGE

```
Step 0. downloadPrecomputedData()*
Step 1. createPairs()
Step 2. computeGenericFeatures()
Step 3. computeCellTypeFeatures()
Step 4. centrePrediction()
```

* This step will be replaced by a supporting AnnotationHub package called CENTREprecomputed that holds the precomputed test data.

CENTRE GitHub

PRECOMPUTED DATA

CENTRE's generic features are computed across many cell types. Generic features provide the potential for an interaction. These statistics are combined test data and crup correlations stored in the PrecomputedData.db database.

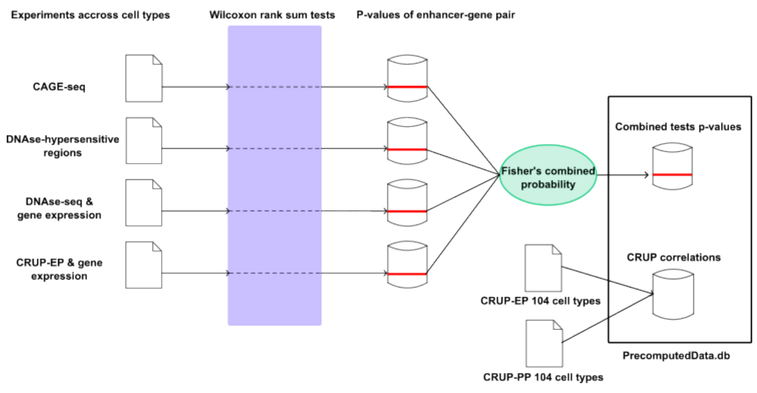

Fig. 2- Outline of PrecomputedData.db. Wilcoxon rank sum tests are computed for each ET pair across many cell types in each of the CAGE-seq[4], DNAse-hypersensitive region[5], DNAse-seq[6], and CRUP expression data sets. The p-values for each ET are combined using Fisher's combined probability[7], which results in a database of combined p-values. Crup correlations are correlations of EP and PP scores for each ET across cell types.

PERFORMANCE AND HI-C IMPROVEMENTS

- CENTRE's performance on the BENGI[8] benchmark is comparable to and, in some cases, better than TargetFinder's.
- CENTRE uses less input data and fewer features than TargetFinder. CENTRE is given as a pre-trained model trained on the consensus LcL dataset [1].
- BENGI is a comprehensive annotation dataset of active and inactive ET pairs across a wide range of cell types.
- All benchmarks were done using a 12-fold cross-validation scheme that ensures data from the same chromosome is not shared between the train and test set to avoid overfitting [8].
- CENTRE's performance is still low. We developed an extension on CENTRE, CENTRE.MI.MSI that uses Hi-C data to provide information about TAD boundaries.
- CENTRE.MI.MSI has two new features: the Minimum Insulation Score and Mean Switch Intensity. The Minimum Insulation Score is based on the Insulation Score[9], and the Mean Switch Intensity is based on the Directionality Index[10] and its sign switches. We plan to add the CENTRE.MI.MSI model to the CENTRE package in the future.

Fig. 3 - (A) Boxplots representing the F1 scores after 12-fold CV achieved by CENTRE and TargetFinder applied in 13 BENGI datasets (B) Comparison of CENTRE and TargetFinder in five tissue datasets where a subset of experimental assays required by TargetFinder was used (TargetFinder Reduced model). The asterisks show the F1 scores of the pre-trained CENTRE classifier (C) F1 scores of the consensus LcL pre-trained CENTRE and CENTRE.MI.MSI models

REFERENCES

[1]Rapakoulia, T., Lopez Ruiz de Vargas, S., Omgba, P. A., Laupert, V., Ulitsky, I., & Vingron, M. (2023). *CENTRE: A gradient boosting algorithm for Cell-type-specific ENhancer-Target pREdiction* (p. 2023.05.16.541035). bioRxiv. https://doi.org/10.1101/2023.05.16.541035

[2]Chen, T., & Guestrin, C. (2016, August). XGBoost. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. https://doi.org/10.1145/2939672.2939785

[3]Ramisch, A., Heinrich, V., Glaser, L. V., Fuchs, A., Yang, X., Benner, P., Schöpflin, R., Li, N., Kinkley, S., Hillmann, A., Longinotto, J., Heyne, S., Czepukojc, B., Kessler, S. M., Kiemer, A. K., Cadenas, C., Arrigoni, L., Gasparoni, N., Manke, T., … Vingron, M. (2019). CRUP: A comprehensive framework to predict condition-specific regulatory units. *Genome Biology*, *20*(1). https://doi.org/10.1186/s13059-019-1860-7

[4]Andersson, R., Gebhard, C., & Miguel-Escalada, I. (2014). An atlas of active enhancers across human cell types and tissues. *Nature*, *507*(7493), 455–461.

[5]Thurman, R. E., Rynes, E., Humbert, R., Vierstra, J., Maurano, M. T., Haugen, E., & al, N. C. S. et. (2012). The accessible chromatin landscape of the human genome. *Nature*, *489*(7414), 75–82. https://doi.org/10.1038/nature11232

[6]Sheffield, N. C., Thurman, R. E., Song, L., Safi, A., Stamatoyannopoulos, J. A., Lenhard, B., Crawford, G. E., & Furey, T. S. (2013). Patterns of regulatory activity across diverse human cell types predict tissue identity, transcription factor binding, and long-range interactions. *Genome Research*, *23*(5), 777–788. https://doi.org/10.1101/gr.152140.112

[7]Fisher, R. A. (1938). *Statistical methods for research workers* (Vol. 23). Protoplasma.

[8]Moore, J. E., Pratt, H. E., Purcaro, M. J., & Weng, Z. (2020). A curated benchmark of enhancer-gene interactions for evaluating enhancer-target gene prediction methods. *Genome Biology*, *21*(1), 17. https://doi.org/10.1186/s13059-019-1924-8

[9]Crane, E., Bian, Q., McCord, R. P., Lajoie, B. R., Wheeler, B. S., Ralston, E. J., Uzawa, S., Dekker, J., & Meyer, B. J. (2015). Condensin-driven remodelling of X chromosome topology during dosage compensation. *Nature*, *523*(7559), Article 7559. https://doi.org/10.1038/nature14450

[10]Dixon, J. R., Selvaraj, S., Yue, F., Kim, A., Li, Y., Shen, Y., Hu, M., Liu, J. S., & Ren, B. (2012). Topological domains in mammalian genomes identified by analysis of chromatin interactions. *Nature*, *485*(7398), Article 7398. https://doi.org/10.1038/nature11082